

The k -Eigenvalue Problem

NE 630

Lesson 14

Prepare: Review Lessons 12–13; bring the multigroup balance equations

Primary objective. Compute the multiplication factor and multigroup spectrum of an infinite, homogeneous system, and distinguish eigenvalue from fixed-source solutions.

1 Keep losses and production separate

Recall $\Sigma_{r,g} = \Sigma_{t,g} - \Sigma_{s,g \leftarrow g}$. For compact algebra, define

$$f_g = (v\Sigma_f)_g, \quad \mathcal{P} = \sum_{g'=1}^G f_{g'} \phi_{g'}.$$

Then the physical steady-state balance is

$$\Sigma_{r,g} \phi_g = \sum_{g' \neq g} \Sigma_{s,g \leftarrow g'} \phi_{g'} + \chi_g \mathcal{P} + s_g.$$

Here f_g is a production cross section in cm^{-1} , while $\mathcal{P}$ is a neutron-production rate density. For the supplied data, $f_g = \bar{v}_g \Sigma_{f,g}$ and $\Sigma_{a,g} = \Sigma_{\gamma,g} + \Sigma_{f,g}$; $\bar{v}_g$ is fission-rate weighted.

2 A multiplying two-group system

Use bounds $(E_0, E_1, E_2) = (10^7, 1, 10^{-3})$ eV. Group 1 is nonthermal and group 2 is thermal. All macroscopic cross sections below are in cm^{-1} .

	$g = 1$	$g = 2$
$\Sigma_{\gamma,g}$	2.04×10^{-3}	1.91×10^{-2}
$\Sigma_{f,g}$	2.96×10^{-4}	1.38×10^{-2}
$\bar{v}_g$	2.49	2.43
$\Sigma_{s,1 \leftarrow g}$	7.21×10^{-1}	0
$\Sigma_{s,2 \leftarrow g}$	4.27×10^{-2}	2.72
v_g (m/s)	10^5	2×10^3

Assume no leakage or out-of-range losses, no upscatter, all fission neutrons born in group 1, and a group-1 external source:

$$(\chi_1, \chi_2) = (1, 0), \quad (s_1, s_2) = (s_1, 0).$$

For numerical fluxes, use $s_1 = 1 \text{ cm}^{-3} \text{ s}^{-1}$.

Reduce the data before solving

Set $d = \Sigma_{s,2 \leftarrow 1} = 0.0427 \text{ cm}^{-1}$. Then

$$\begin{aligned} \Sigma_{a,1} &= 0.002336, & \Sigma_{a,2} &= 0.0329, \\ \Sigma_{r,1} &= \Sigma_{a,1} + d = 0.045036, \\ f_1 &= 0.00073704, & f_2 &= 0.033534. \end{aligned}$$

All of these coefficients have units cm^{-1} . Why is $\Sigma_{r,2} = \Sigma_{a,2}$ here?

3 Solve the fixed-source problem

The two balances are

$$\begin{aligned} \Sigma_{r,1} \phi_1 &= f_1 \phi_1 + f_2 \phi_2 + s_1, \\ \Sigma_{a,2} \phi_2 &= d \phi_1. \end{aligned}$$

First the shape, then the magnitude

The thermal equation determines the flux ratio:

$$R = \frac{\phi_2}{\phi_1} = \frac{d}{\Sigma_{a,2}} = \underline{\hspace{2cm}}.$$

Insert $\phi_2 = R\phi_1$ into the first balance:

$$\phi_1 = \frac{s_1}{\Sigma_{r,1} - f_1 - f_2 R}, \quad \phi_2 = R\phi_1.$$

The denominator is $0.00077610894 \text{ cm}^{-1}$. For the stated unit source,

$$\begin{aligned} \phi_1 &= \underline{\hspace{2cm}} \text{ cm}^{-2} \text{ s}^{-1}, \\ \phi_2 &= \underline{\hspace{2cm}} \text{ cm}^{-2} \text{ s}^{-1}. \end{aligned}$$

For another source strength, the fluxes scale in direct proportion to s_1 while the material data remain fixed.

4 Audit the solution

Adding the group balances cancels intergroup scattering:

$$\Sigma_{a,1} \phi_1 + \Sigma_{a,2} \phi_2 = f_1 \phi_1 + f_2 \phi_2 + s_1.$$

Using unrounded fluxes, the rates in $\text{cm}^{-3} \text{ s}^{-1}$ are

$$\begin{aligned} R_a &= \underline{\hspace{2cm}}, \\ \mathcal{P} &= \underline{\hspace{2cm}}, \\ R_a - \mathcal{P} &= \underline{\hspace{2cm}}. \end{aligned}$$

Why were speeds supplied?

Speeds do not enter these steady-state flux balances. Using v_g as a representative group speed gives $n_g \approx \phi_g / v_g$. Convert the tabulated speeds to cm/s , then compare n_2/n_1 with ϕ_2/ϕ_1 .

$$\frac{n_2}{n_1} \approx \frac{\phi_2}{\phi_1} \frac{v_1}{v_2} = \underline{\hspace{2cm}}.$$

5 Remove the source; introduce k

With $s_1 = 0$, a nonzero *physical steady-state* solution requires exactly critical material. Instead, define an eigenproblem by dividing **only fission production** by k_∞ :

$$\Sigma_{r,1}\phi_1 = \frac{1}{k_\infty}(f_1\phi_1 + f_2\phi_2),$$

$$\Sigma_{a,2}\phi_2 = d\phi_1.$$

The second equation is unchanged, so $R = d/\Sigma_{a,2}$.

Derive the eigenvalue

Divide the first equation by $\phi_1 > 0$:

$$k_\infty \Sigma_{r,1} = f_1 + f_2 R.$$

$$k_\infty = \frac{f_1 + f_2 d / \Sigma_{a,2}}{\Sigma_{r,1}} = \underline{\hspace{2cm}}.$$

The unmodified, source-free system is therefore
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The $1/k_\infty$ factor is mathematical normalization of production, not a physical source or a time-growth rate.

6 Write the matrix eigenproblem

Define the loss-minus-scattering and fission matrices:

$$\mathbf{M} = \begin{bmatrix} \Sigma_{r,1} & 0 \\ -d & \Sigma_{a,2} \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} f_1 & f_2 \\ 0 & 0 \end{bmatrix}.$$

Fixed source versus eigenvalue

$$(\mathbf{M} - \mathbf{F})\phi = \mathbf{s} \quad (\text{fixed source}),$$

$$\mathbf{M}\phi = \frac{1}{k_\infty} \mathbf{F}\phi \quad (\text{eigenvalue}).$$

When $\mathbf{M}$ is invertible,

$$\mathbf{M}^{-1}\mathbf{F}\phi = k_\infty \phi.$$

For G groups the same structure uses

$$M_{gg'} = \Sigma_{t,g}\delta_{gg'} - \Sigma_{s,g \leftarrow g'}, \quad F_{gg'} = \chi_g f_{g'}.$$

Checkpoint

Why is the off-diagonal entry $-d$ in $\mathbf{M}$ negative, although the scattering cross section itself is positive?

7 Normalize a shape, not a power

An eigenvector has arbitrary magnitude. One convenient convention is

$$\hat{\phi}_g = \frac{\phi_g}{\phi_1 + \phi_2}, \quad \hat{\phi}_1 + \hat{\phi}_2 = 1.$$

Report the eigen-spectrum

$$\hat{\phi}_1 = \frac{1}{1 + R} = \underline{\hspace{2cm}},$$

$$\hat{\phi}_2 = \frac{R}{1 + R} = \underline{\hspace{2cm}}.$$

These are dimensionless fractions of total flux, not neutron number fractions. What additional information would set a physical flux magnitude?

For complete energy coverage, summing the eigen-equations gives

$$k_\infty = \frac{\sum_g f_g \phi_g}{\sum_g \Sigma_{a,g} \phi_g}.$$

Evaluate this with the *eigen-spectrum*; it recovers the one-group production-to-absorption ratio from Lesson 12.

8 Connect to source multiplication

For this model, the external source and fission births both enter group 1. Therefore the fixed-source and eigenvalue problems have the same flux ratio R . Using $f_1 + f_2 R = k_\infty \Sigma_{r,1}$ in Section 3 yields

$$\phi_1 = \frac{s_1}{\Sigma_{r,1}(1 - k_\infty)}, \quad \phi_2 = R\phi_1.$$

If fission *production* is suppressed while keeping the same loss and scattering coefficients, the group-1 flux is $\phi_1^{(0)} = s_1/\Sigma_{r,1}$. Hence

$$\frac{\phi_1}{\phi_1^{(0)}} = \frac{1}{1 - k_\infty} = \underline{\hspace{2cm}}.$$

Approach criticality from below

At fixed $s_1 > 0$, what happens as $k_\infty \rightarrow 1^-$? Why is the negative algebraic flux obtained for $k_\infty > 1$ not a physical steady-state solution?

Takeaway

A fixed source sets the flux magnitude in a subcritical system. The eigenproblem sets k_∞ and a relative spectrum, not power. The simple $1/(1 - k_\infty)$ factor here relies on this source shape and fixed two-group model; it is not universal.